



ELK406E

Distribution of Electrical Energy

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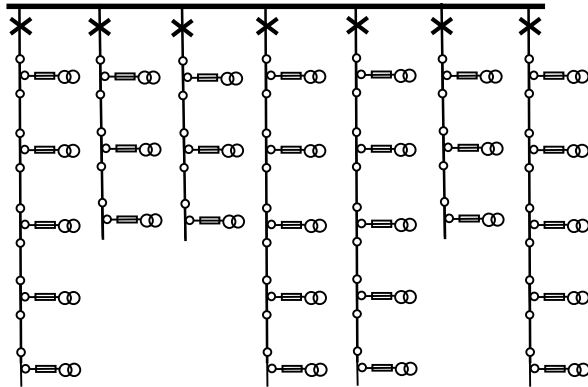
Primary Distribution

- There are several configurations of primary distribution systems.
 - Radial system
 - Loop system
 - Open loop system
 - Closed loop system
 - Network system

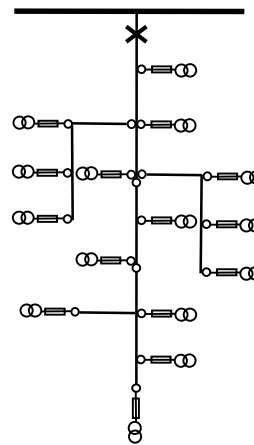
Primary Distribution

radial system

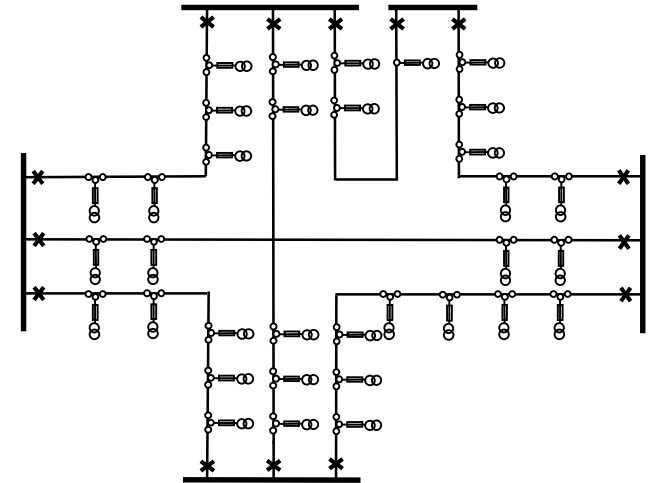
Urban



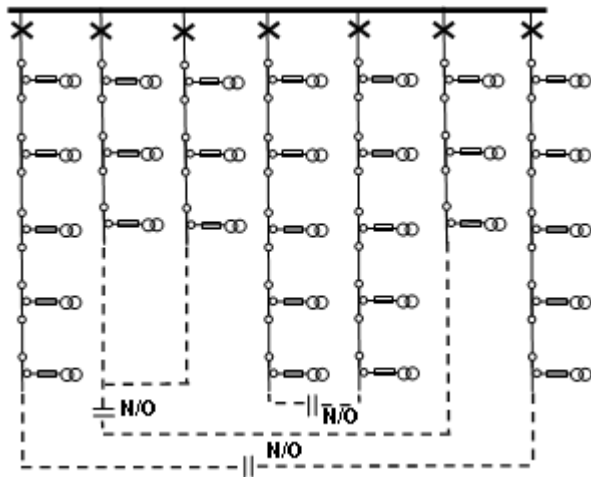
Rural



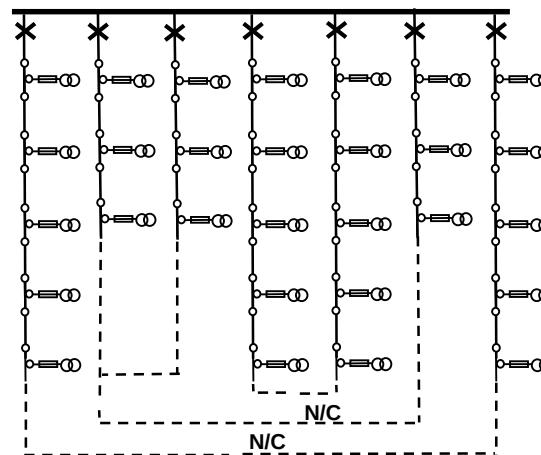
Primary Network system



Open loop system



Closed loop system



Medium Voltage
Distribution
Network
Structures

Primary Distribution

The primary-feeder voltage level is the most important factor affecting the system design, cost, and operation. Some of the design and operation aspects affected by the primary-feeder voltage level are [1]:

1. Primary-feeder length
2. Primary-feeder loading
3. Number of distribution substations
4. Rating of distribution substations
5. Number of subtransmission lines
6. Number of customers affected by a specific outage
7. System maintenance practices
8. The extent of tree trimming
9. Joint use of utility poles
10. Type of pole-line design and construction
11. Appearance of the pole line.

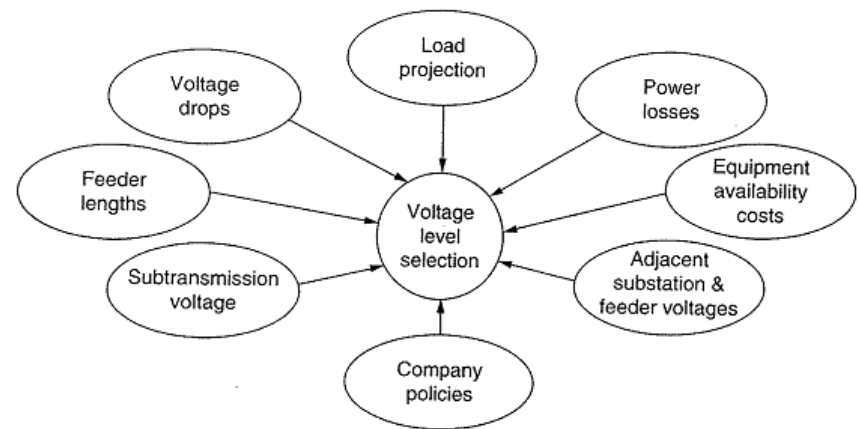


FIGURE 5.8 Factors affecting primary-feeder voltage-level selection decision.

- In general, for a given percent voltage drop, the feeder length and loading are direct functions of the feeder voltage level.
- This relationship is known as the voltage-square rule.
- For example, if the feeder voltage is doubled, for the same percent voltage drop, it can supply the same power four times the distance.

TABLE 5.1
Typical Primary Voltage Levels

Class	3 ϕ Voltage	
2.5 kV	2300	3W- Δ
	2400*	3 W- Δ
5.0 kV	4000	3W- Δ or 3W-Y
	4160*	4W-Y
	4330	3W- Δ
	4400	3W- Δ
	4600	3W- Δ
	4800	3W- Δ
8.66 kV	6600	3W- Δ
	6900	3W- Δ or 4W-Y
	7200*	3W- Δ or 4W-Y
	7500	4W-Y
	8320	4W-Y
	11000	3W- Δ
15 kV	11500	3W- Δ
	12000	3W- Δ or 4W-Y
	12470*	4W-Y
	13200*	3W- Δ or 4W-Y
	13800*	3W- Δ
	14400	3W- Δ
25 kV	22900*	4W-Y
	24940*	4W-Y
34.5 kV	34500*	4W-Y

* Most common voltage in the individual classes.

Primary-feeder loading is defined as the loading of a feeder during peak load conditions as measured at the substation [1]. Some of the factors affecting the design loading of a feeder are:

1. The density of the feeder load
2. The nature of the feeder load
3. The growth rate of the feeder load
4. The reserve capacity requirements for emergency
5. The service continuity requirements
6. The service reliability requirements
7. The quality of service
8. The primary-feeder voltage level
9. The type and cost of construction
10. The location and capacity of the distribution substation
11. The voltage regulation requirements

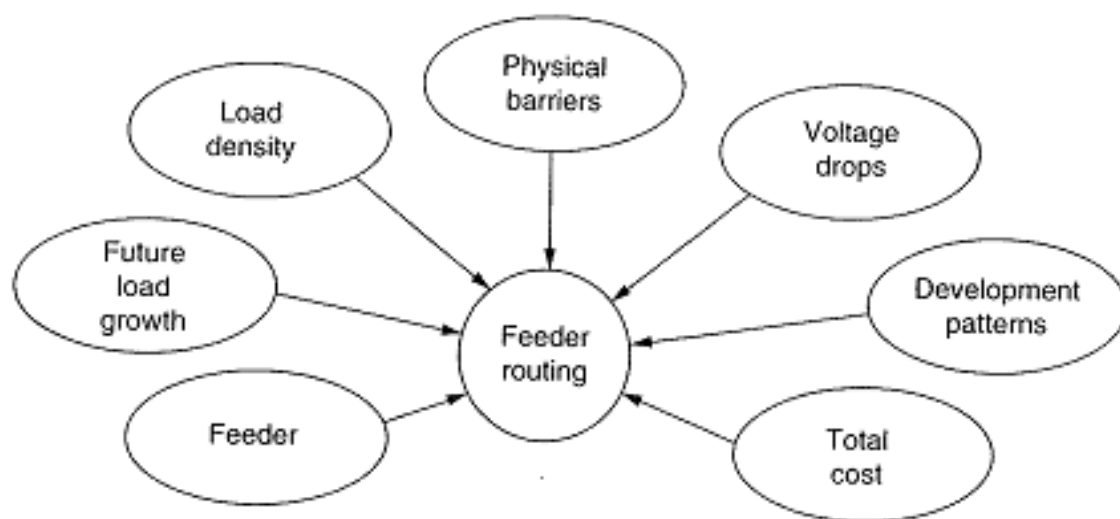


FIGURE 5.11 Factors affecting feeder routing decisions.

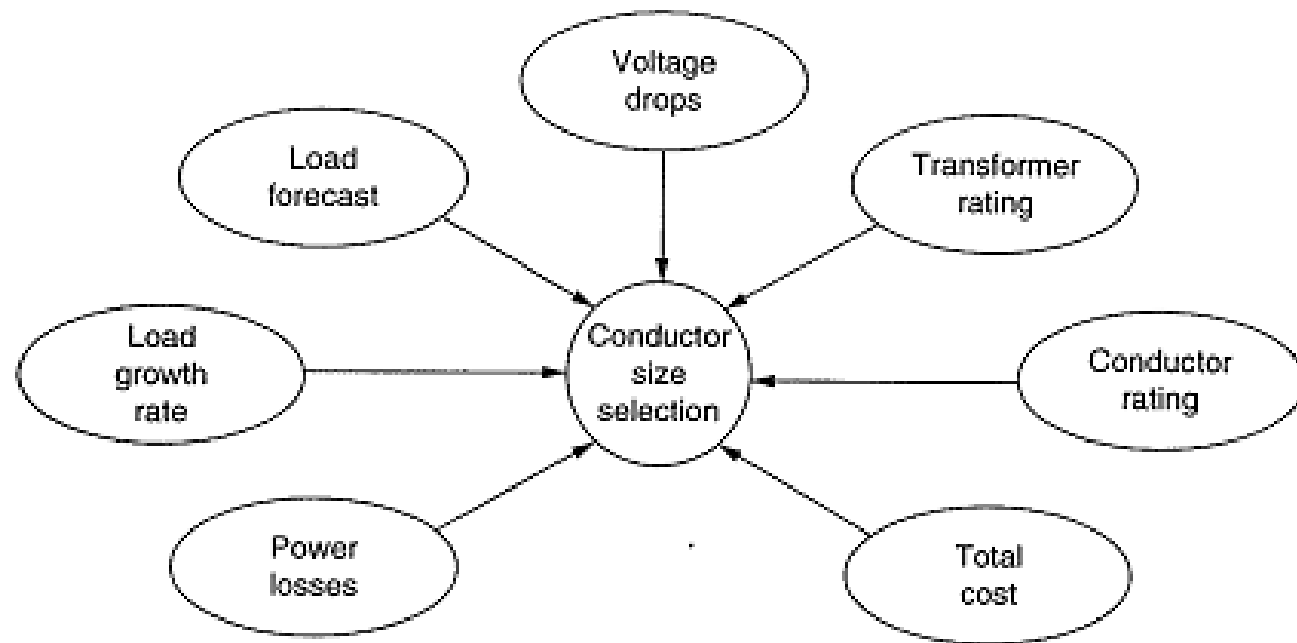


FIGURE 5.13 Factors affecting conductor size selection.

A MV substation usually contains the following major components;

- Transformers
- Circuit Breakers
- Air-Break switches
- Disconnect switches
- Grounding switches
- Surge arresters
- Current-limiting reactors
- Instrument transformers
- Relay and protective devices
- Sensors, panels etc

Circuit Breakers

- CBs are designed to interrupt either normal or short-circuit currents
- They behave like big switches that may be opened or closed by local pushbuttons or by distant telecommunication signals emitted by the system of protection

The most important types of CBs are the following:

- Oil-CBs
- Air-blast CBs
- SF6 CBs
- Vacuum CBs



Vacuum CB

Air-Break Switches

- Can not interrupt normal load currents
- Can interrupt exciting currents of TRs
- Can interrupt moderate capacitive currents of unloaded lines

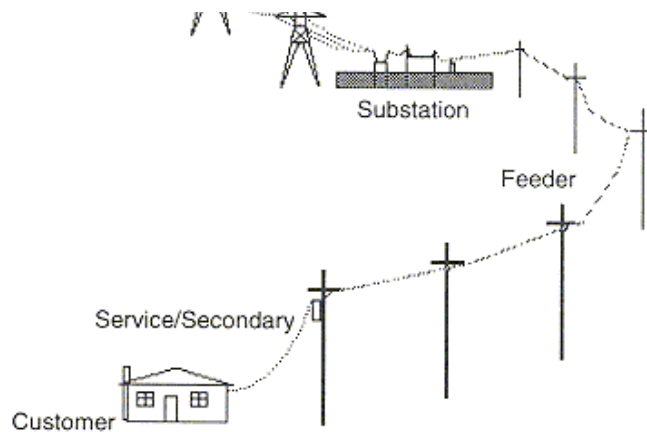


Disconnecting Switches

Unable to interrupt any current

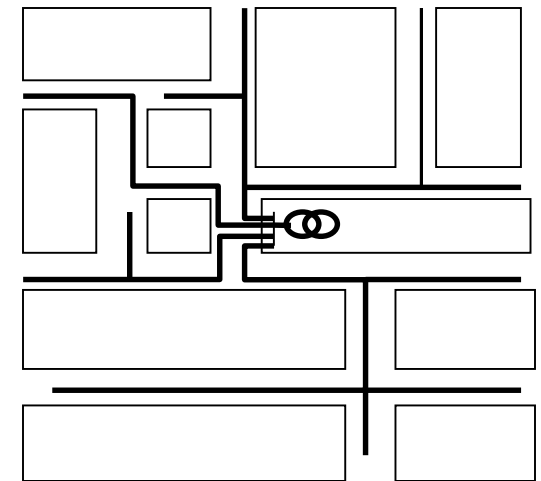


Secondary Distribution Systems



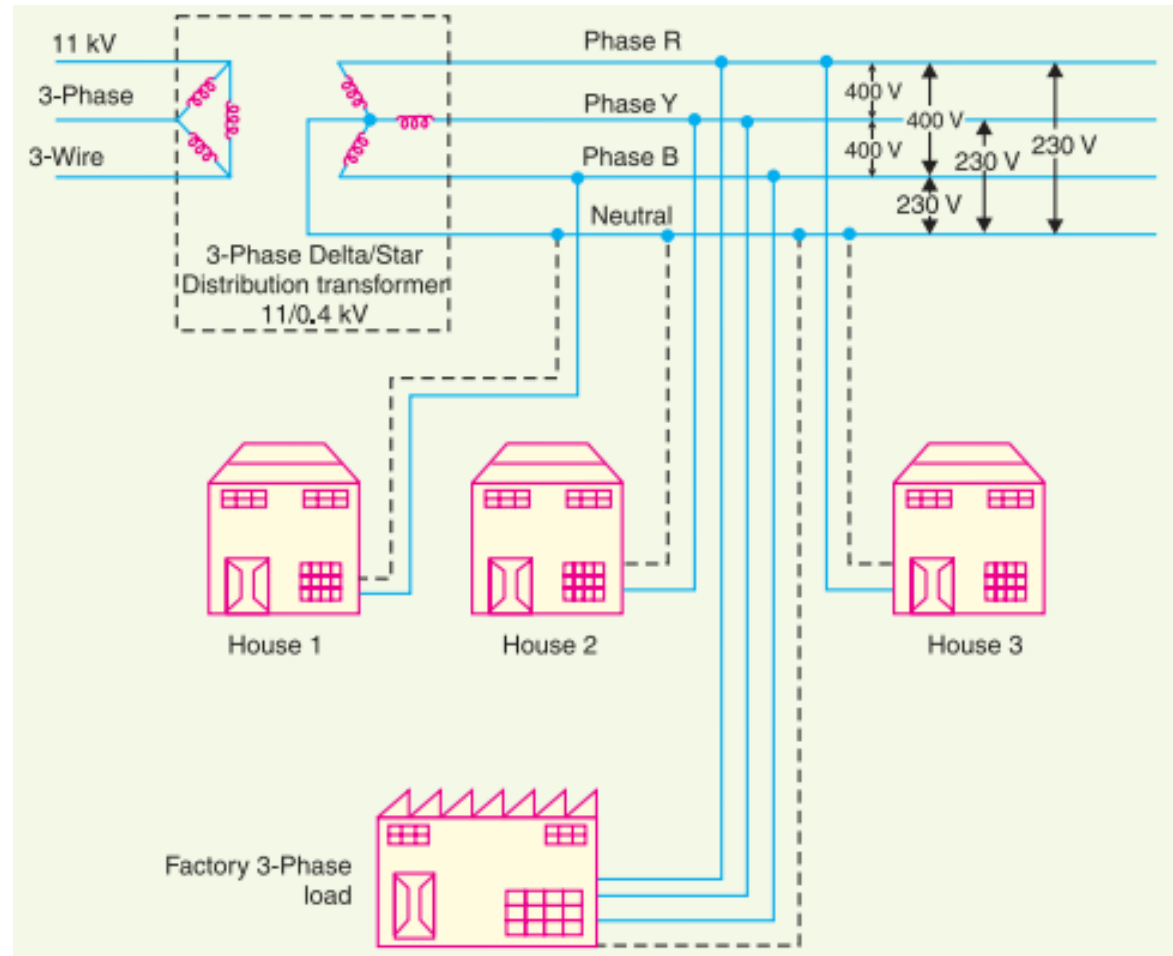
- The secondary distribution system is that part of the electrical power system between the primary system and the customer's service entrance.
- This system includes distribution transformers, secondary circuits (secondary mains), customer services (consumer drops), and watt-hour meters to measure customer power consumption.
- Secondary voltages are provided by distribution transformers that are connected to the primary system and sized for the voltages required for specific parts of the service area

The secondary distribution systems are generally of the radial type except for some specific service areas such as hospitals, business centers and military installations, which require highly reliable service, and therefore, may be of a grid or mesh type



- Heavy industries or mines, which require the most power, are usually supplied with three-phase power by privately owned or corporate industrial substations.
- Factories, high-rise buildings, shopping centers, and other large power consumers are furnished with three-phase power from load substations

- Many commercial and light industrial customers are supplied by 400/230 V, three-phase, four-wire systems



To provide maximum reliability and operating flexibility, utilities use spot and grid network systems in congested area such as metropolitan and suburban business districts.

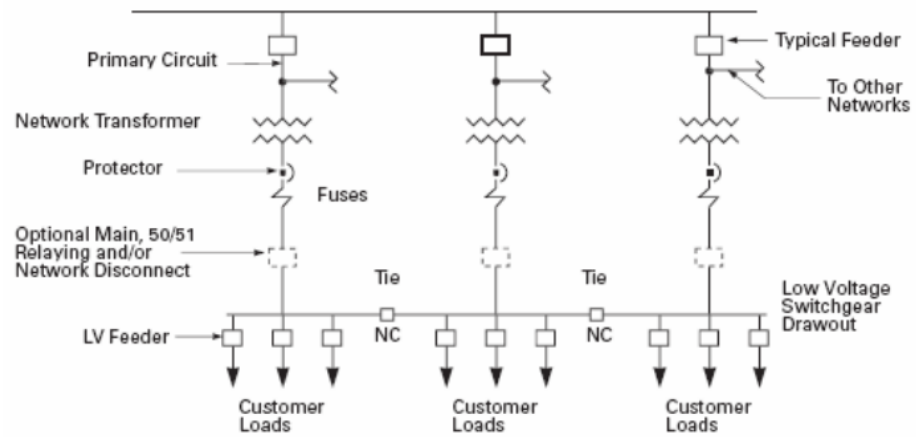


Figure 5. Spot network system

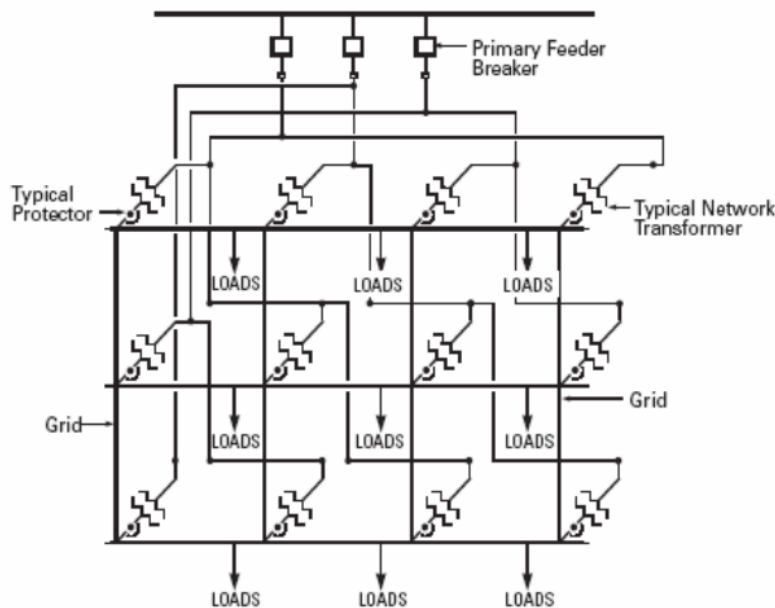


Figure 6. Secondary grid network system
(one primary source shown; may have two or more)

Spot network: A secondary network distribution system that consists of two or more network units at a single site.

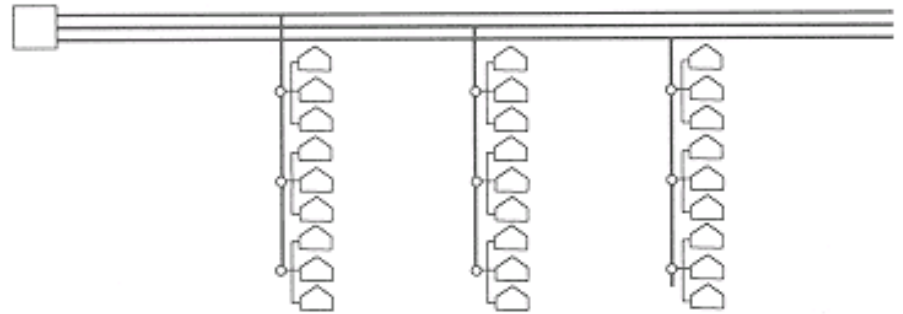
Secondary network distribution system

An AC power distribution system in which customers are served from three-phase, four-wire low-voltage circuits supplied by two or more network transformers whose low-voltage terminals are connected to the low-voltage circuits through network protectors.

•Secondary Distribution

In USA

Secondary distribution 120 V, 1 ϕ is used.
Each transformer (1 ϕ) supplies 4-20 house.
In Rural Areas, generally, pole-mounted transformers are used.

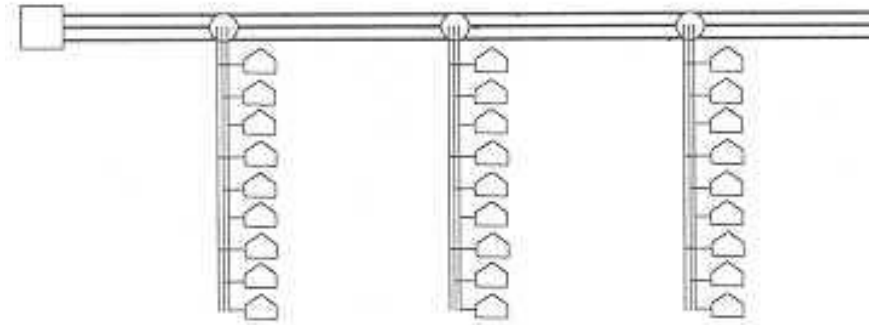


In Turkey

400 V, 3 ϕ is used.

In general, transformers (3 ϕ) supply a region.

- Transformers (<400 kVA) can be pole-mounted.
- Transformers (>400 kVA) are in the transformer building



Relative to North American designs, European systems have larger transformers and more customers per transformer. Most European transformers are three-phase and on the order of 300 to 1000 kVA, much larger than typical North American 25- or 50-kVA singlephase units

North America has standardized on a 120/240-V secondary system;
on these, voltage drop constrains how far utilities can run secondaries, typically no more than 250 ft.

In European designs, higher secondary voltages allow secondaries to stretch to almost 1 mi.

European secondaries are largely three-phase and most European countries have a standard secondary voltage of 220, 230, or 240 V.

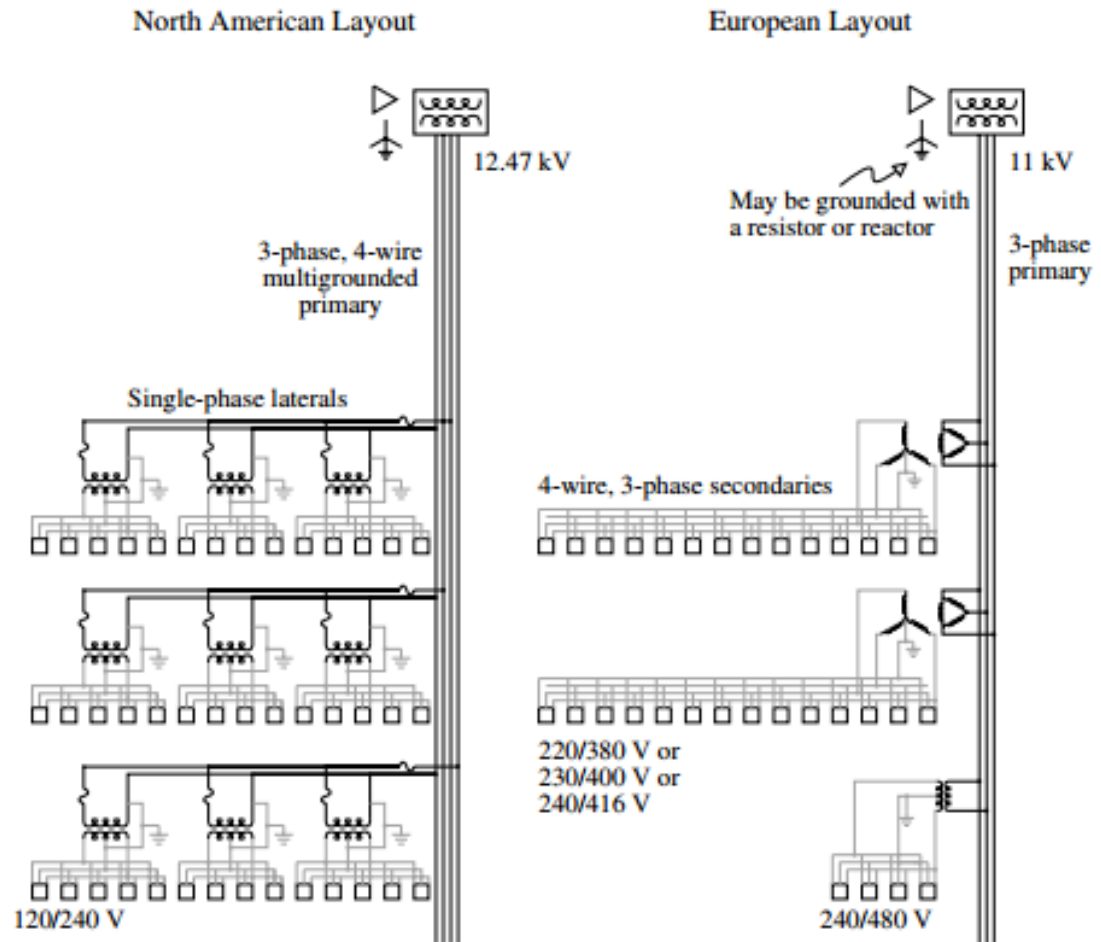


FIGURE 1.17
North American versus European distribution layouts.

Some of the major differences between systems

Cost — The European system is generally more expensive than the North American system

Flexibility — The North American system has a more flexible primary design, and the European system has a more flexible secondary design.

Reliability — Generally, North American designs result in fewer customer interruptions.

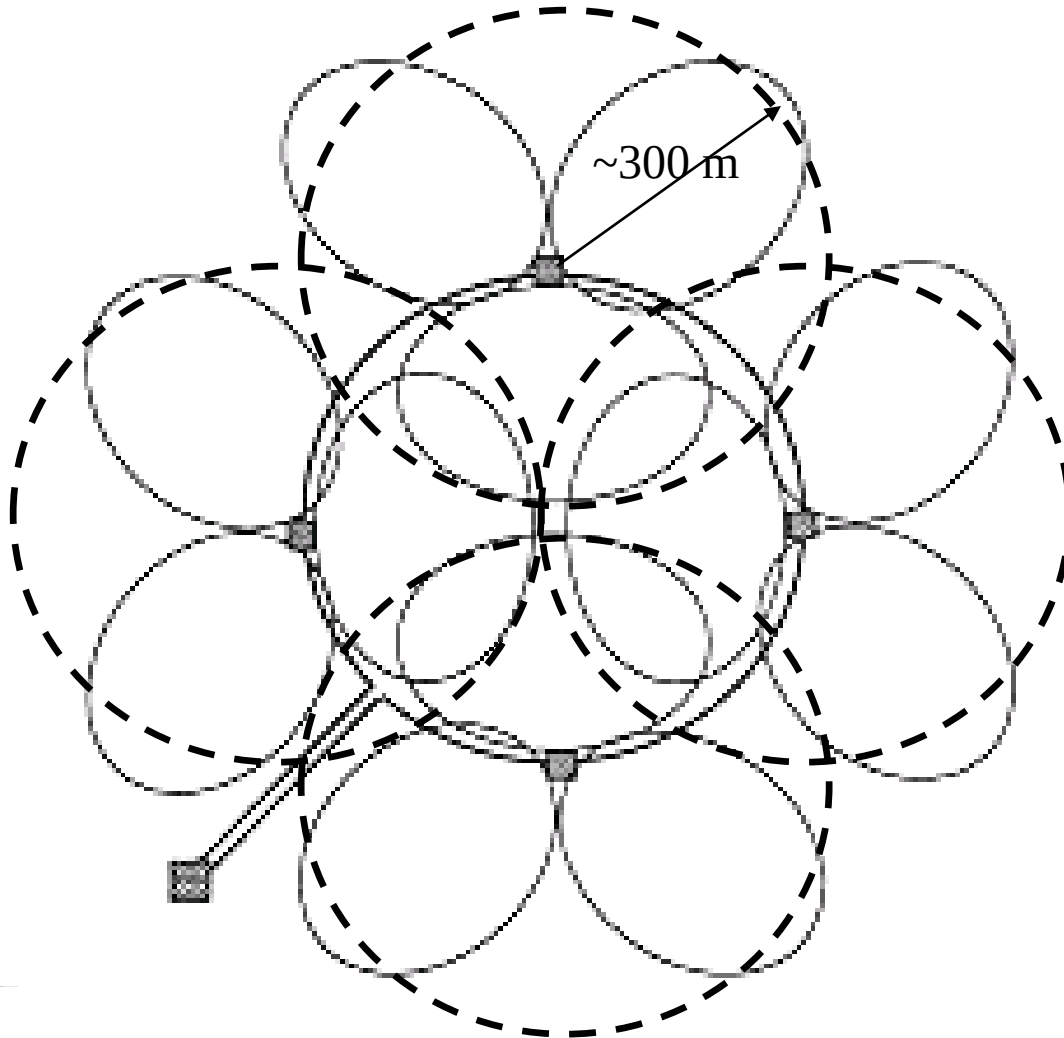
Safety — The multigrounded neutral of the North American primary system provides many safety benefits; protection can more reliably clear faults, and the neutral acts as a physical barrier, as well as helping to prevent dangerous touch voltages during faults. The European system has the advantage that high-impedance faults are easier to detect

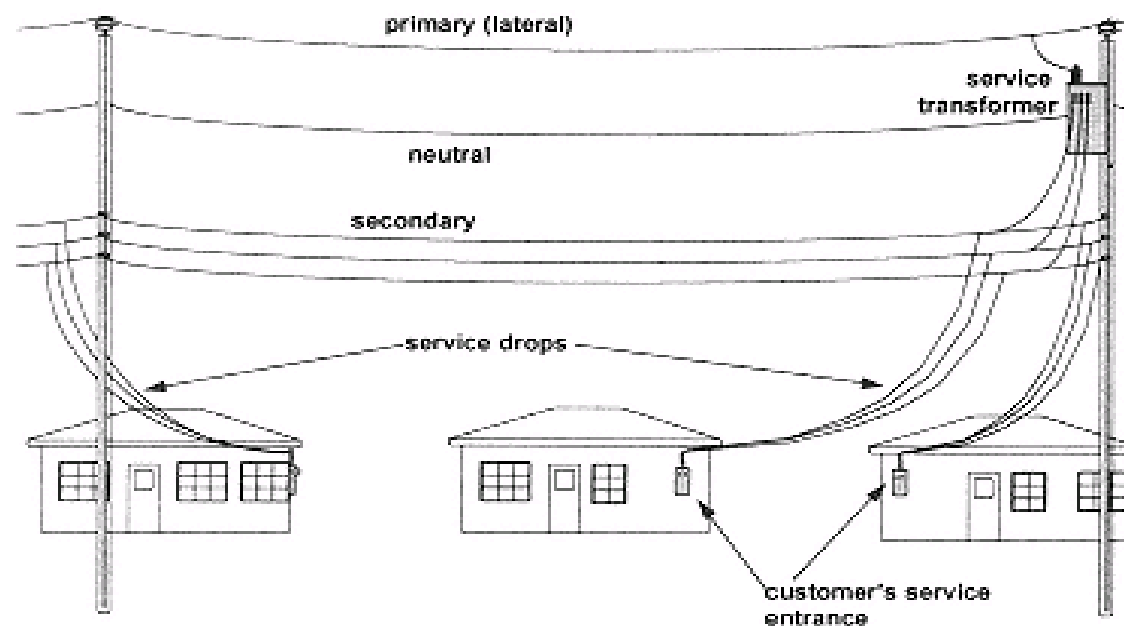
Power quality — Generally, European systems have fewer voltage sags and momentary interruptions.

Aesthetics — Having less primary, the European system has an aesthetic advantage: the secondary is easier to underground or to blend in. For underground systems, fewer transformer locations and longer secondary reach make siting

Theft — The flexibility of the European secondary system makes power much easier to steal. Developing countries especially have this problem. Secondaries are often strung along or on top of buildings; this easy access does not require great skill to attach into.

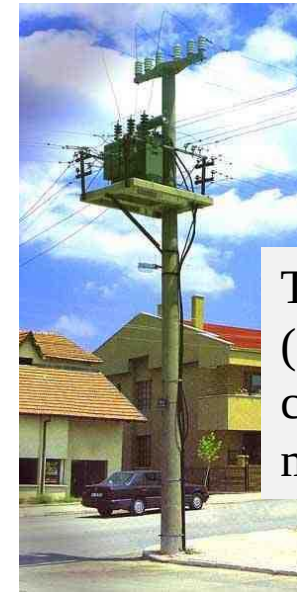
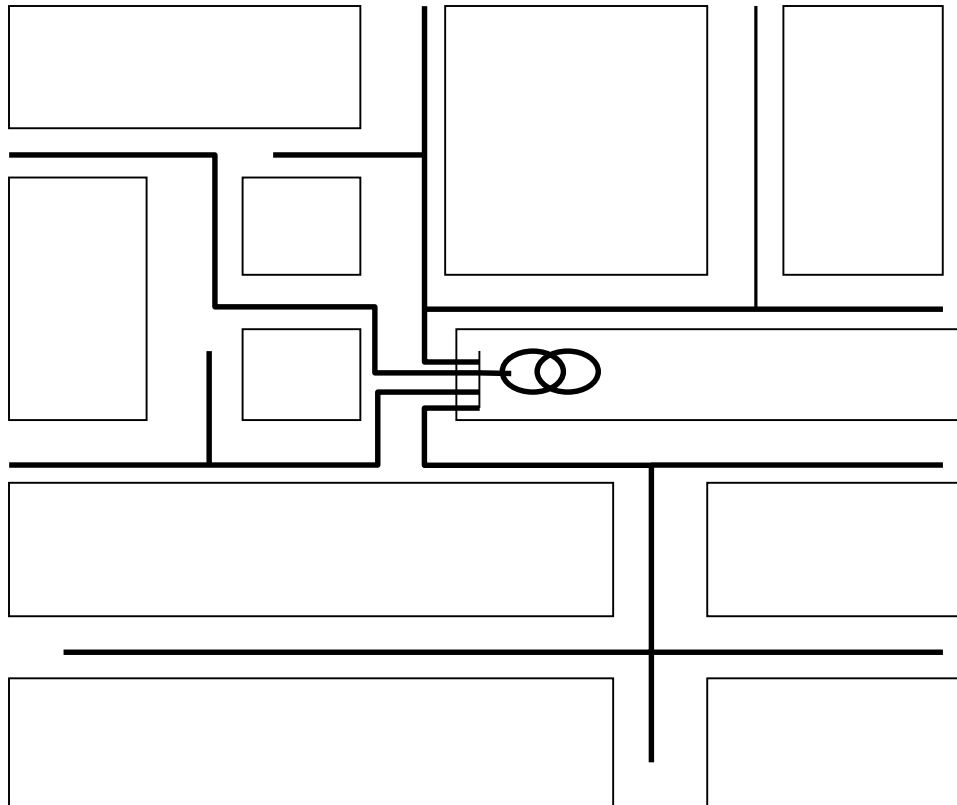
In general, transformers (3ϕ) supply a region which has approximately 200-300 m radius





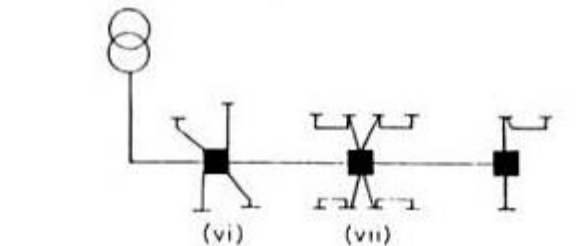
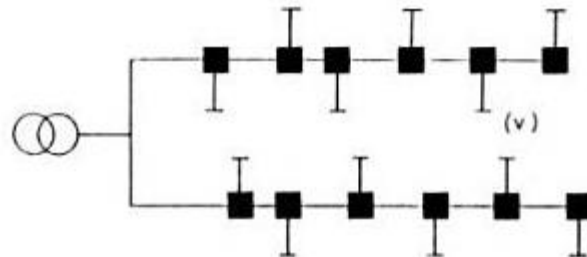
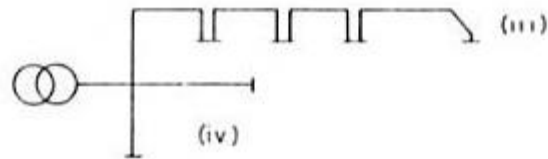
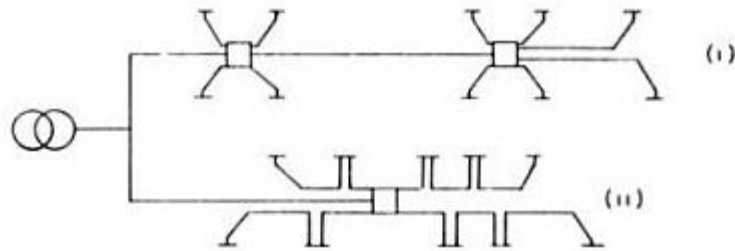
The service level, often the most visible to the average electric power consumer, consists of service transformers, secondary circuits, and service drops. Shown here is a typical overhead 120/240 volt (American) type of system fed from a single-phase lateral.

- The low voltage distribution system distributes the electric energy from the secondary distribution substations to the low voltage customers.
- The voltage (LV) is below 1 kV



Transformers (<400 kVA) can be pole-mounted.

In developed areas, the distribution is done via underground cables.



In arrangement (i) the disconnection boxes not only provide facilities to connect up the services but also make it possible to provide fuses on each outgoing cable. Arrangements (ii) and (iii) are cheaper owing to the linking connections, but do not permit such individual good protection facilities as arrangement (i). Arrangement (iv) is for individual large or remote customers. Arrangement (v) represents the situation with fixed underground joints, which are much cheaper than cable disconnection boxes or cabinets although the selective protection facility discussed earlier is not possible.

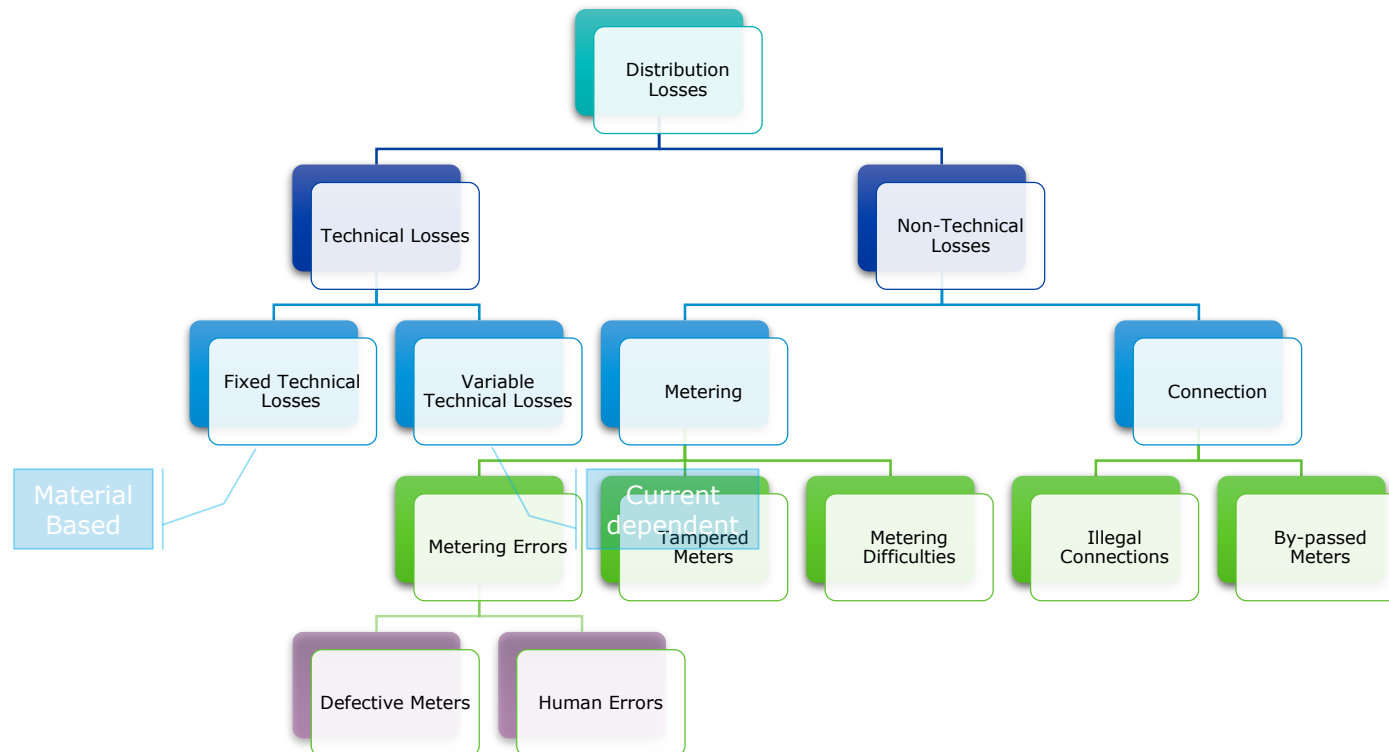
Arrangements (vi) and (vii) are variations on arrangement (v). In particular, arrangement (vii) has proved to be cost effective in countries where each customer in a terraced house has his or her own service and this simple arrangement meets the local safety regulations.

e 10.11 Some service cable arrangements

- disconnection box
- customer
- underground fixed joint

Non-Technical Losses

- Distribution Losses: Difference between the infeed energy and total billed energy at the end customer side



smart meters



With the advent of smart meters,

- the frequency of collecting household energy consumption data has increased,
- The advanced data analysis is possible

Smart meter data and data from distribution transformer can be used to detect electricity theft in an area



Designing of a new distribution system includes following parts;

for Primary distribution

- Determination of total power demand
- Decision of
 - voltage level
 - line type (overhead or underground)
 - system type (radial or loop)
- Selection of routes
- Determination of conductor types and cross-sections
by using following calculations
 - voltage drop
 - power loss
 - short circuit
 - current carrying capacity
- Determination of poles (If the system is overhead)

For secondary distribution

- To find the best possible location for the transformer
- Decision and designing of secondary distribution lines (overhead or underground)
- Design of street lighting
- Selection of all primary and secondary system devices (protection, metering etc.)
- Calculation of system costs (Cost/Benefit Analysis)
- Design of poles (If the system is overhead)

An overhead distribution line consists of

- conductors,
- insulators,
- support structures

Conductors

Virtually all overhead lines use bare aluminum conductors because of their economy, good electrical conduction properties and light weight.

- An all-steel conductor is seldom used because it does not conduct electricity efficiently.
- Copper is slightly better than aluminum.
- Aluminum is lighter, a decided advantage because it allows for less expensive poles and towers.
- The most popular type of conductor is called ACSR (Aluminum Clad, Steel Reinforced), made of aluminum strands, wound around the outside of an inner core of several steel strands.
- The aluminum conducts electricity very well, while the steel provides high strength.

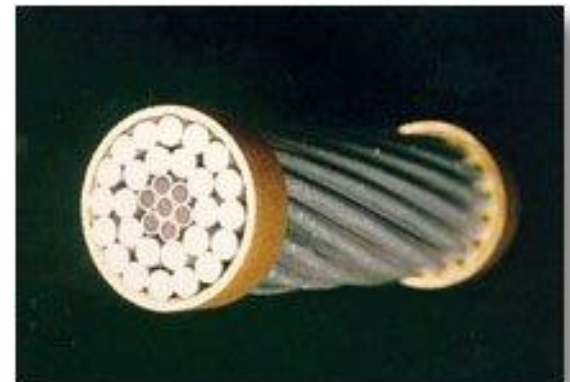
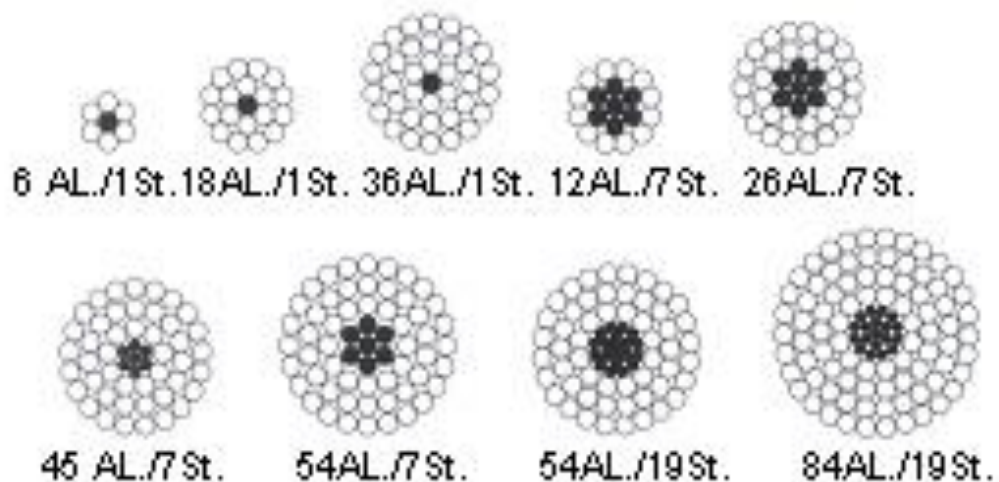


One of the most common conductor types is *aluminum conductor, steelreinforced (ACSR)*, which consists of layers of aluminum strands surrounding a central core of steel strands.

Each layer of strands is spiraled in the opposite direction of its adjacent layer.

Stranded conductors are also easier to handle and more flexible than solid conductors, especially in larger sizes.

The use of steel strands gives ACSR conductors a high strength-to-weight ratio.



For purposes of heat dissipation, overhead line conductors are bare (no insulating cover).

Other conductor types include

AAC All aluminum conductor

AAAC All aluminum alloy conductor

ACAR Aluminum conductor alloy reinforced



All aluminium conductor (AAC) is used mainly in urban areas and the spacing is short and the supports are close. They are made up of one or more strands of aluminium wire depending on the end usage.



ACSR conductor consists of a solid or stranded steel core surrounded by strands of aluminium.



ACSR is available in a wide range of steel varying from as low as 6% to as high as 40 %.

The higher strength ACSR is used for river crossings, overhead ground wires, installations involving extra long spans etc..

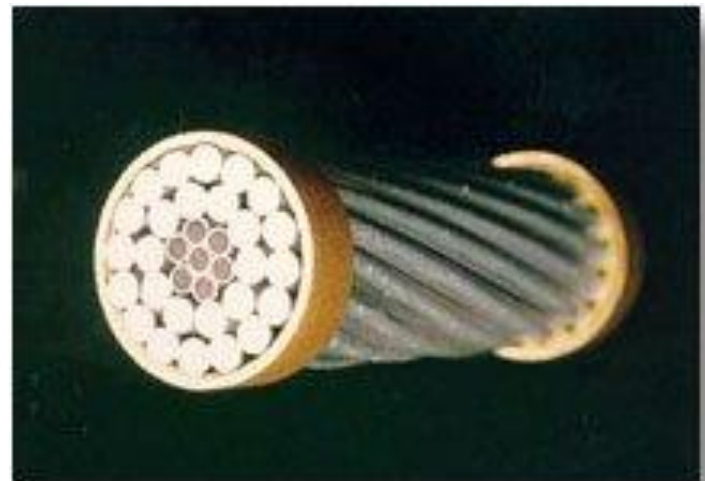


Table 14-7 Aluminum Conductor—Physical Characteristics EC-H19 Classes AA and A

Cable code word	Conductor size		Current-carrying capacity* A	Stranding		Conductor diam, in	Rated strength, lb	Nominal weight, lb†	
	cmils or AWG	in ²		Class	No. and diam of wires, in			Per 1000 ft	Per mile
Peachbell	6	0.0206	95	A	7 × 0.0612	0.184	563	24.6	130
Rose	4	0.0328	130	A	7 × 0.0772	0.232	881	39.2	207
Iris	2	0.0522	175	AA, A	7 × 0.0974	0.292	1,350	62.3	329
Pansy	1	0.0657	200	AA, A	7 × 0.1093	0.328	1,640	78.5	414
Poppy	1/0	0.0829	235	AA, A	7 × 0.1228	0.368	1,990	99.1	523
Aster	2/0	0.1045	270	AA, A	7 × 0.1379	0.414	2,510	124.9	659
Phlox	3/0	0.1317	315	AA, A	7 × 0.1548	0.464	3,040	157.5	832
Oxlip	4/0	0.1663	365	AA, A	7 × 0.1739	0.522	3,830	198.7	1,049
Sneezewort	250,000	0.1964	405	AA	7 × 0.1890	0.567	4,520	234.7	1,239
Valerian	250,000	0.1963	405	A	19 × 0.1147	0.574	4,660	234.6	1,239
Daisy	266,800	0.2097	420	AA	7 × 0.1953	0.586	4,830	250.6	1,323
Laurel	266,800	0.2095	425	A	19 × 0.1185	0.593	4,970	250.4	1,322
Peony	300,000	0.2358	455	A	19 × 0.1257	0.629	5,480	281.8	1,488
Tulip	336,400	0.2644	495	A	19 × 0.1331	0.666	6,150	316.0	1,668
Daffodil	350,000	0.2748	506	A	19 × 0.1357	0.679	6,390	328.4	1,734
Canna	397,500	0.3124	550	AA, A	19 × 0.1447	0.724	7,110	373.4	1,972
Goldentuft	450,000	0.3534	545	AA	19 × 0.1539	0.770	7,890	422.4	2,230
Cosmos	477,000	0.3744	615	AA	19 × 0.1584	0.793	8,360	447.5	2,363
Syringa	477,000	0.3743	615	A	37 × 0.1135	0.795	8,690	447.4	2,362
Zinnia	500,000	0.3926	635	AA	19 × 0.1622	0.811	8,760	469.2	2,477
Hyacinth	500,000	0.3924	635	A	37 × 0.1162	0.813	9,110	469.0	2,476
Dahlia	556,500	0.4369	680	AA	19 × 0.1711	0.856	9,750	522.1	2,757
Mistletoe	556,500	0.4368	680	AA, A	37 × 0.1226	0.858	9,940	522.0	2,756
Meadowsweet	600,000	0.4709	715	AA, A	37 × 0.1273	0.891	10,700	562.8	2,972
Orchid	636,000	0.4995	745	AA, A	37 × 0.1311	0.918	11,400	596.9	3,152
Heuchera	650,000	0.5102	755	AA	37 × 0.1325	0.928	11,600	609.8	3,220
Verbena	700,000	0.5494	790	AA	37 × 0.1375	0.963	12,500	656.6	3,467
Flag	700,000	0.5495	790	A	61 × 0.1071	0.964	12,900	656.8	3,468
Violet	715,500	0.5622	800	AA	37 × 0.1391	0.974	12,800	672.0	3,548
Nasturtium	715,500	0.5619	800	A	61 × 0.1083	0.975	13,100	671.6	3,546
Petunia	750,000	0.5892	825	AA	37 × 0.1424	0.997	13,100	704.3	3,719
Cattail	750,000	0.5892	825	A	61 × 0.1109	0.998	13,500	704.2	3,718
Arbutus	795,000	0.6245	855	AA	37 × 0.1466	1.026	13,900	746.4	3,941
Lilac	795,000	0.6248	855	A	61 × 0.1142	1.028	14,300	746.7	3,943

Table 14-8 ACSR Conductor—Physical Characteristics

ACSR				Current-carrying capacity,* A	Stranding No. and diam of strand, in		Diameter, in		Nominal weight, lb†			Rated strength, lb			
Cross section			Per 1000 ft						Zinc-coated core						
Aluminum		Total	Standard weight coating						Class B coating	Class C coating	Aluminum-coated core				
Code word	cmils or AWG	in ²			in ²										
Turkey	6	0.0206	0.0240	95	6 × 0.0661	1 × 0.0661	0.198	0.0661	36.1	24.5	11.6	1,190	1,160	1,120	1,120
Swan	4	0.0328	0.0382	130	6 × 0.0834	1 × 0.0834	0.250	0.0834	57.4	39.0	18.4	1,860	1,810	1,760	1,760
Swanate	4	0.0328	0.0411	130	7 × 0.0772	1 × 0.1029	0.257	0.1029	67.0	39.0	28.0	2,360	2,280	2,200	2,160
Sparrow	2	0.0522	0.0608	175	6 × 0.1052	1 × 0.1052	0.316	0.1052	91.3	62.0	29.3	2,850	2,760	2,680	2,640
Sparate	2	0.0522	0.0654	175	7 × 0.0974	1 × 0.1299	0.325	0.1299	106.7	62.0	44.7	3,640	3,510	3,390	3,260
Robin	1	0.0657	0.0767	200	6 × 0.1181	1 × 0.1181	0.355	0.1181	115.1	78.2	36.9	3,550	3,450	3,340	3,290
Raven	1/0	0.0830	0.0968	230	6 × 0.1327	1 × 0.1327	0.398	0.1327	145.3	98.7	46.6	4,380	4,250	4,120	3,980
Quail	2/0	0.1046	0.1221	265	6 × 0.1490	1 × 0.1490	0.447	0.1490	183.2	124.4	58.8	5,310	5,130	5,050	4,720
Pigeon	3/0	0.1317	0.1537	310	6 × 0.1672	1 × 0.1672	0.502	0.1672	230.8	156.7	74.1	6,620	6,410	6,300	5,880
Penguin	4/0	0.1662	0.1939	350	6 × 0.1878	1 × 0.1878	0.563	0.1878	291.1	197.7	93.4	8,350	8,080	7,950	7,420
Waywing	266,800	0.2094	0.2210	430	18 × 0.1217	1 × 0.1217	0.609	0.1217	289.5	250.3	39.2	6,880	6,770	6,650	6,540
Owl	266,800	0.2096	0.2368	410	6 × 0.2109	7 × 0.0703	0.633	0.2109	342.4	250.5	91.9	9,680	9,420	9,160	9,160
Partridge	266,800	0.2095	0.2436	440	26 × 0.1013	7 × 0.0788	0.642	0.2364	367.3	251.7	115.6	11,300	11,000	10,600	10,640
Merlin	336,400	0.2642	0.2789	500	18 × 0.1367	1 × 0.1367	0.684	0.1367	365.2	315.7	49.5	8,680	8,540	8,400	8,260
Linnet	336,400	0.2640	0.3070	510	26 × 0.1137	7 × 0.0884	0.720	0.2652	462.5	317.0	145.5	14,100	13,700	13,300	13,300
Oriole	336,400	0.2642	0.3259	515	30 × 0.1059	7 × 0.1059	0.741	0.3177	527.1	318.1	209.0	17,300	16,700	16,200	15,900
Chickadee	397,500	0.3121	0.3295	555	18 × 0.1486	1 × 0.1486	0.743	0.1486	431.6	373.1	58.5	9,940	9,780	9,690	9,530
Brant	397,500	0.3122	0.3527	565	24 × 0.1287	7 × 0.0858	0.772	0.2574	512.1	375.0	137.1	14,600	14,300	13,900	13,900
Ibis	397,500	0.3119	0.3627	570	26 × 0.1236	7 × 0.0961	0.783	0.2883	546.6	374.7	171.9	16,300	15,800	15,300	15,100
Lark	397,500	0.3121	0.3849	575	30 × 0.1151	7 × 0.1151	0.806	0.3453	622.7	375.8	246.9	20,300	19,600	18,900	18,600
Pelican	477,000	0.3747	0.3955	625	18 × 0.1628	1 × 0.1628	0.814	0.1628	518.0	447.8	70.2	11,800	11,600	11,500	11,100
Flicker	477,000	0.3747	0.4233	635	24 × 0.1410	1 × 0.1628	0.846	0.2820	614.6	450.1	164.5	17,200	16,700	16,200	16,000

Insulators

The live conductors are insulated from the towers by insulators which take two basic forms: **the pin type and the suspension type**. It is seen that the conductor is supported on the top of the pin type-insulator. This type is used for lines up to 33 kV.

Suspension-type insulators consist of a string of discs, typically porcelain. **The number of insulator discs in a string increases with line voltage.**



The selection of cost-effective and technically feasible solutions to achieve the required transmission system reliability improvements and to interconnect the specified substations and switching stations

The avoidance, minimization, or mitigation of adverse environmental, cultural, and economic effects

Selection of routes

principles for transmission & distribution overhead line route selection;

- select the shortest route
- parallel highways, roads, rail roads as much as possible
- follow section lines
- route in direction of possible future loads
- avoid crossing hills, ridges, swamps and bottom lands
- avoid paralleling telephone lines
- preserve the environment



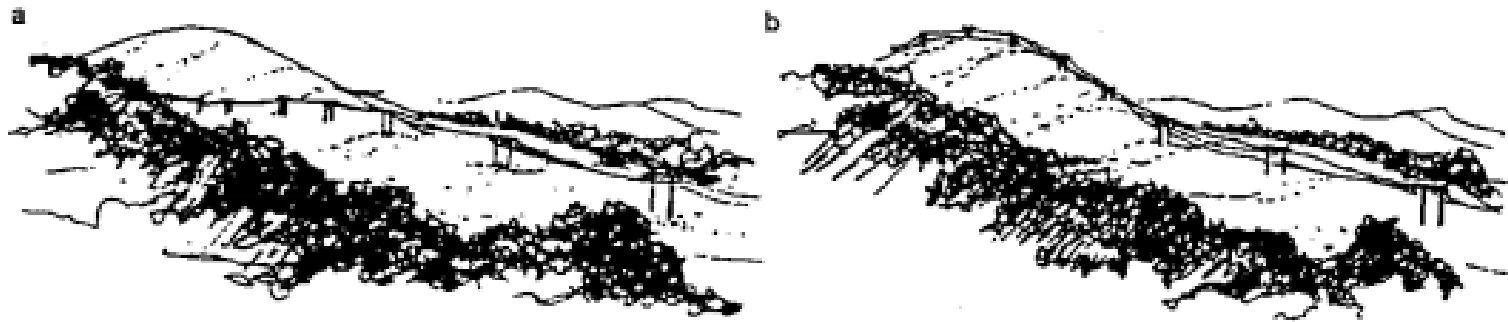


Figure 15.1: Line routing to avoid breaking the skyline. a recommended; b to be avoided

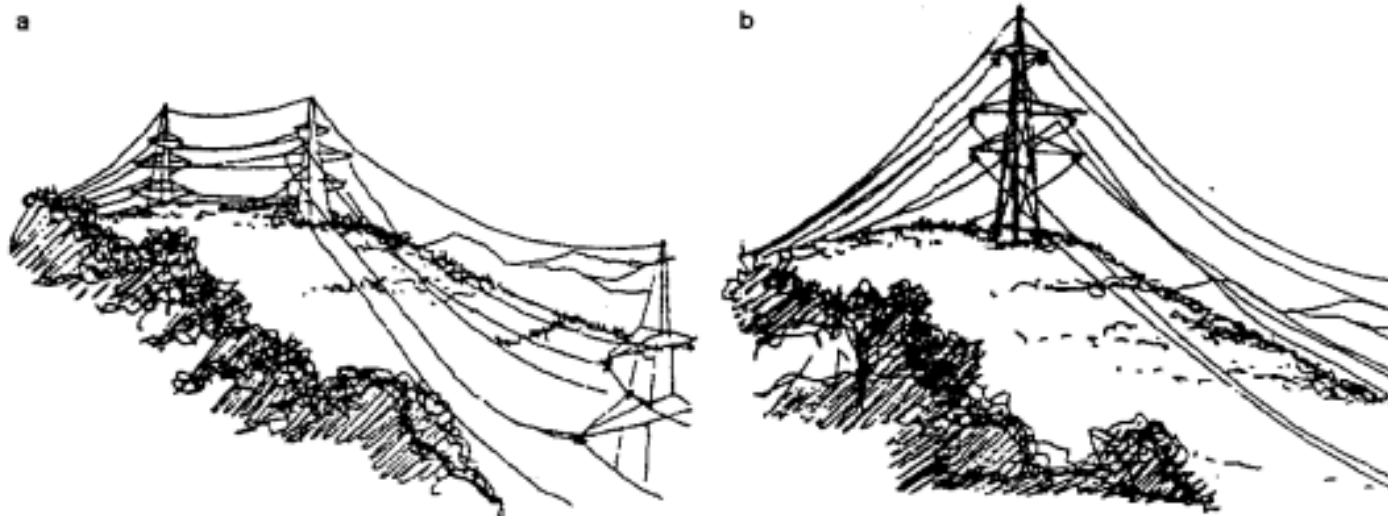


Figure 15.2: Line routing on hills. a recommended; b to be avoided

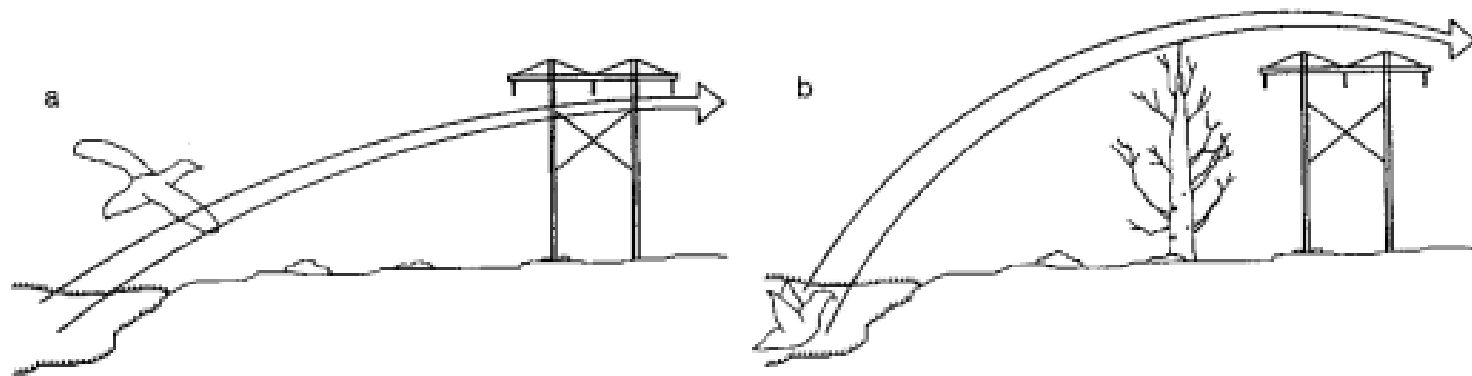
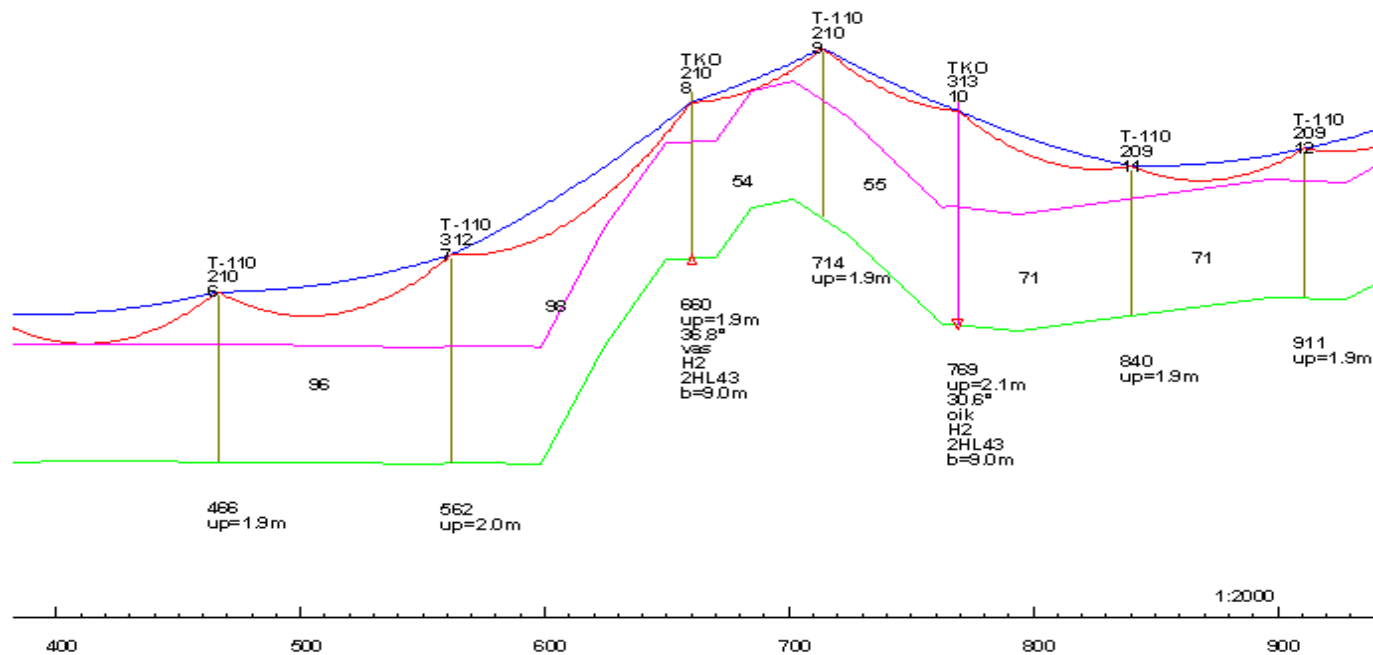


Figure 15.3: Line routing to avoid collision with birds (according to [15.5]). a not desirable, b to aspire

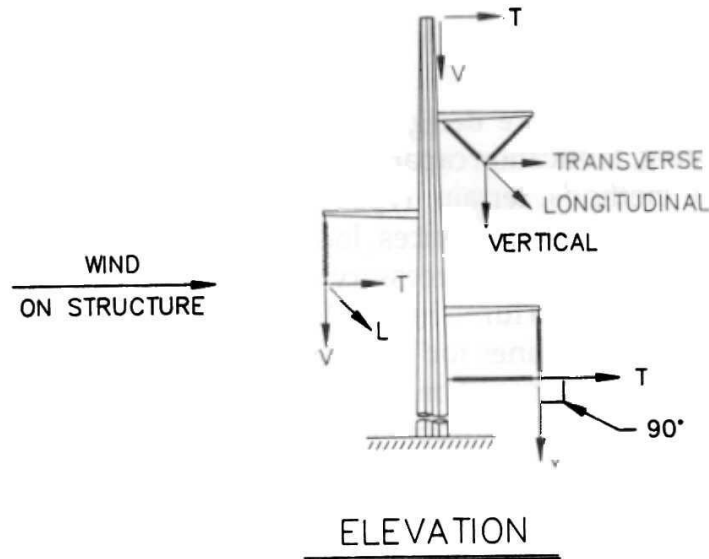
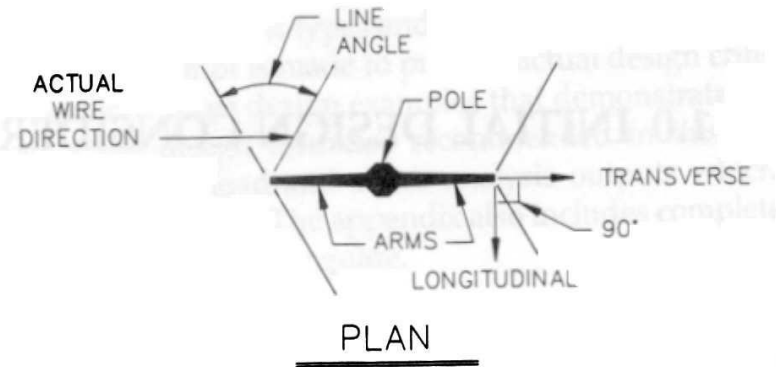


survey data in both the plan and the profile coordinate systems



The forces on a pole stem from the vertical, horizontal (transverse and longitudinal) loading.

The forces on the pole are the dead weight of the conductors, ice loading, wind force, the weight of arms, insulators, transformers etc.

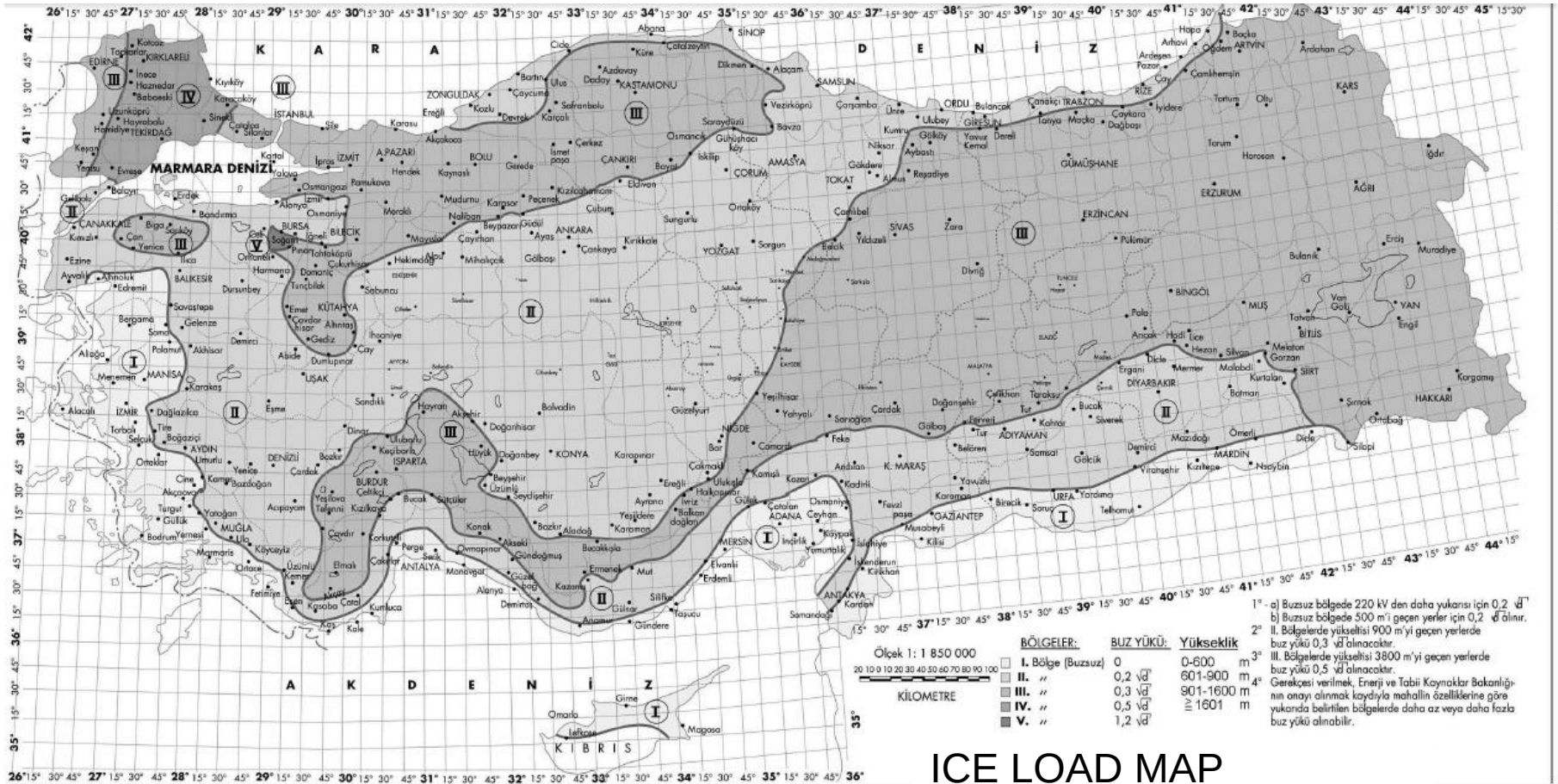




The whether related loads on the overhead distribution and transmission system are

- wind velocity
- ice thickness
- temperature

The NESC divides the USA into three loading areas.

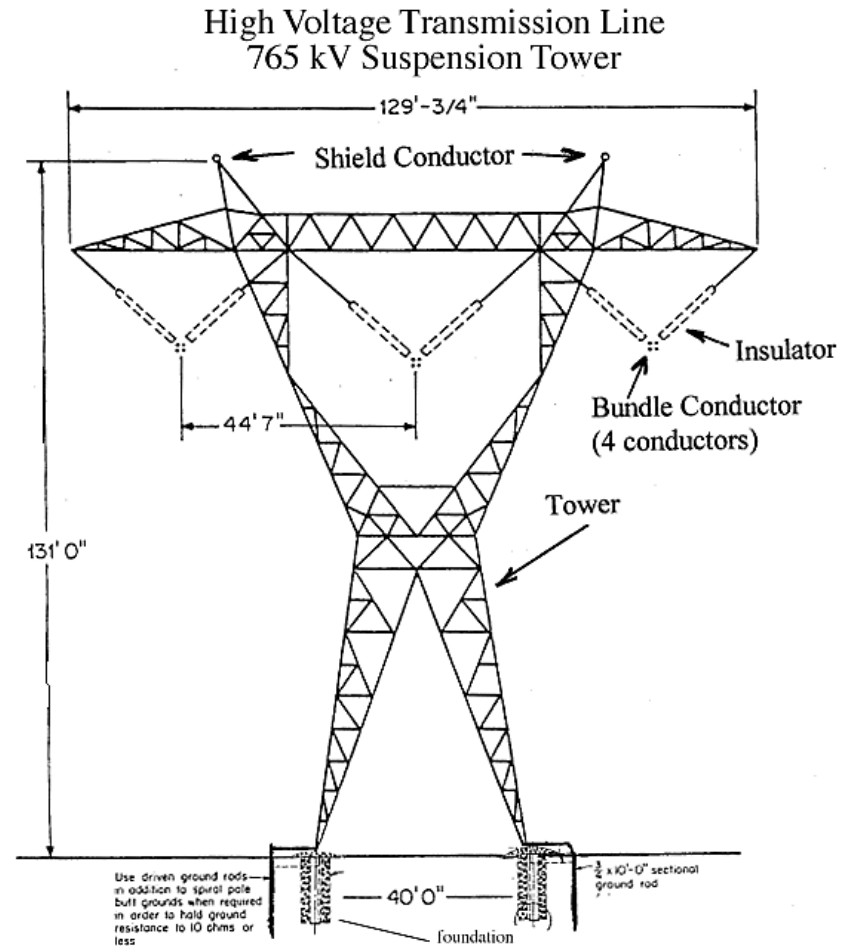


ICE LOAD MAP

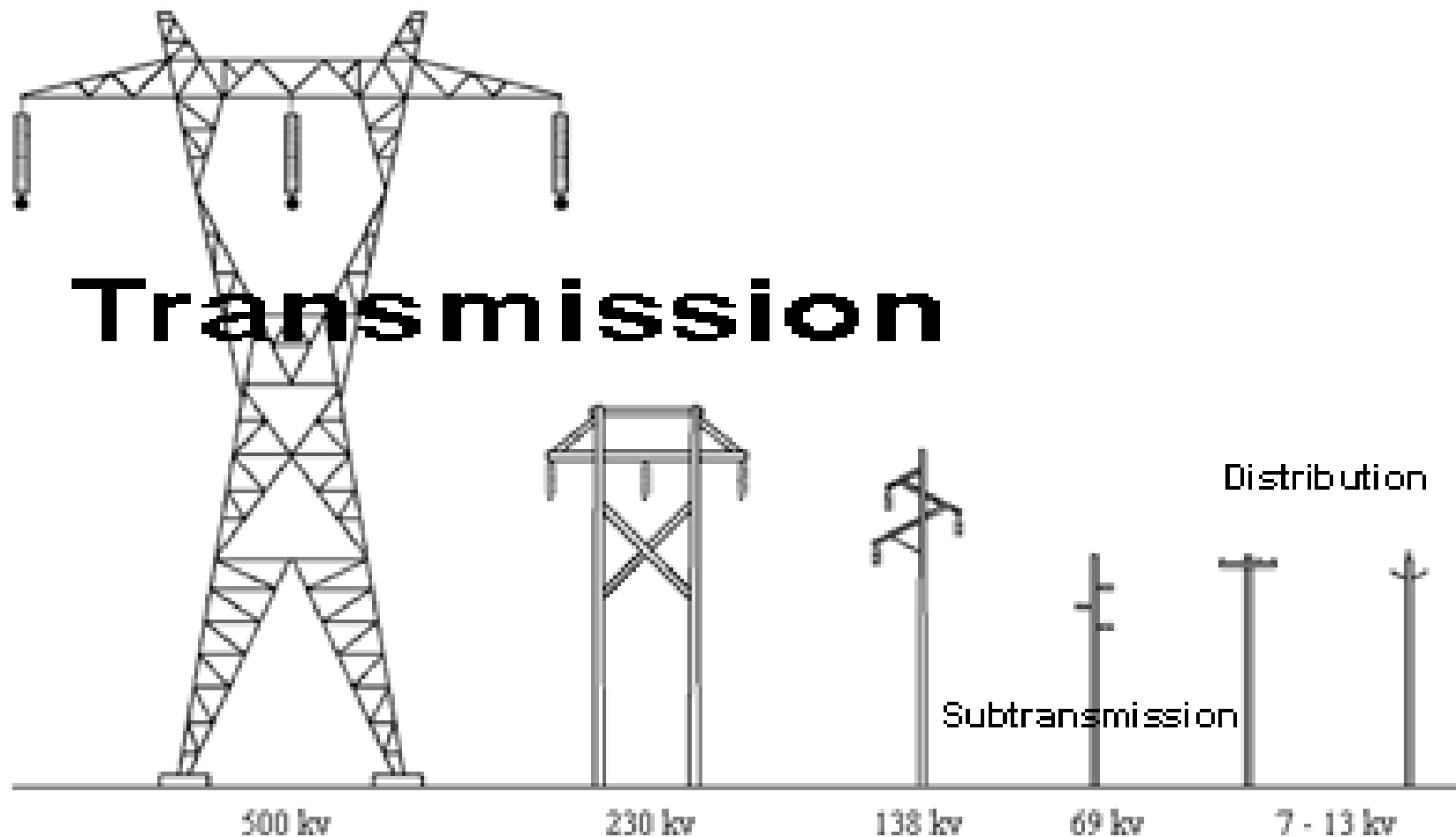
Support Structures

These serve the purpose of keeping the conductors at a safe height from ground as well as at an adequate distance from each other.

The construction of the support is dependent on the cost. The cost takes into account the design and the materials as well as transportation and labor.



Transmission



When specifying towers and lines, ice and wind loadings are taken into account, as well as extra forces due to a break in the lines on one side of a tower.



Pole Structures are

- wood poles

common used in USA

tree types are cedar, pine



- concrete poles

greater cost, longer life

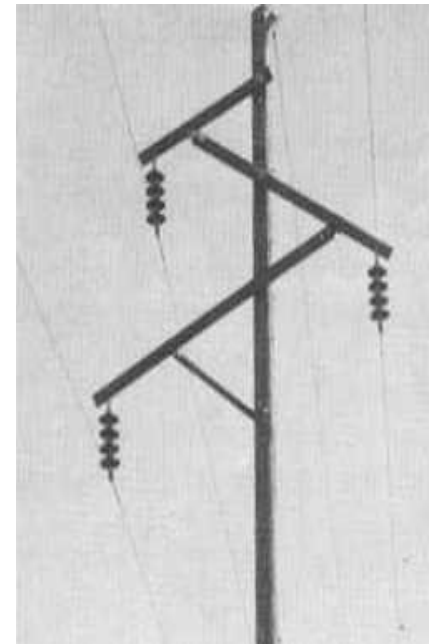
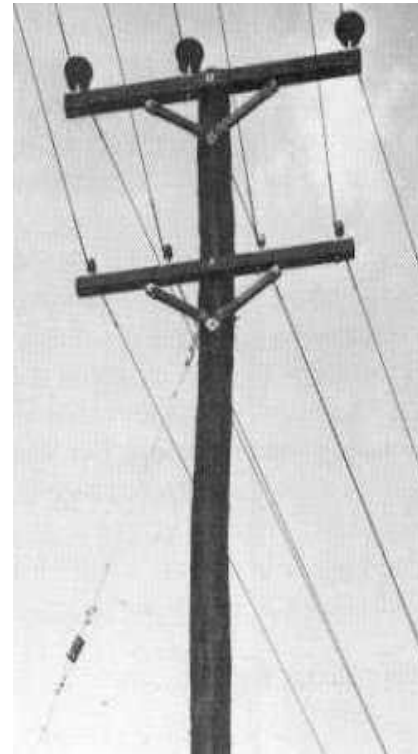
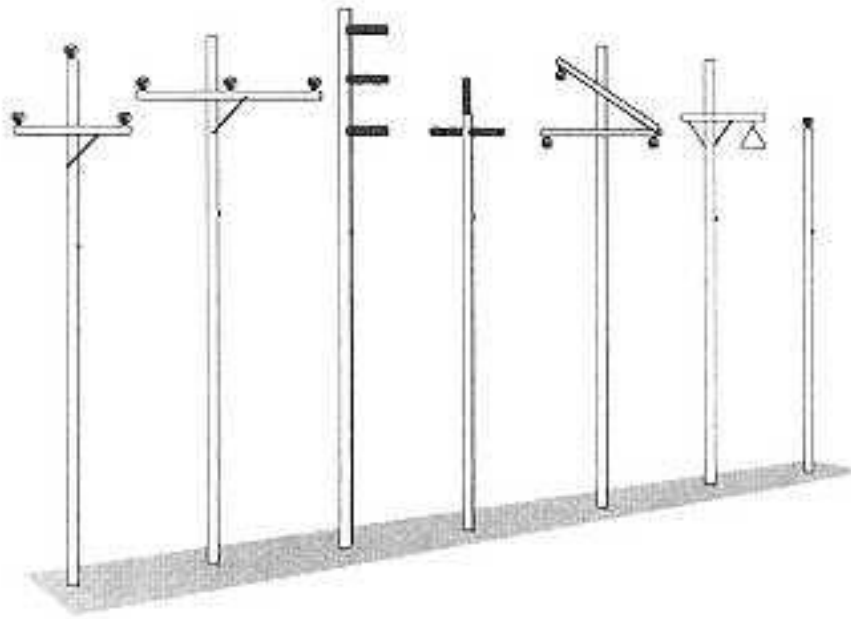
greater strength

- metal poles

- steel poles

- aluminium poles



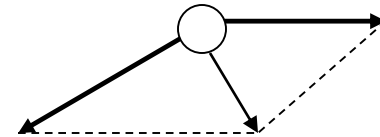


All poles are divided into classes based on thickness and circumference (strength)

Classification of Poles

To simplify the analysis, poles have been classified as:

- Straightaway poles (in a straight line)
transverse loading
- Angle poles (at angles or offsets in the line)
tensions, angle loading, transverse loading



- Dead-end poles

Dead-end poles have different tensions or conductor sizes on opposite sides of the pole. This creates an intact unbalanced longitudinal load.

GUYS

When the horizontal loads imposed on poles and cross arms may exceed the safe carrying strengths of the wood involved (or the holding power of the soil), guys of steel wire are usually installed to take up or counteract the excessive stress.

The guys take up the horizontal stresses and distribute them to other cross arms, to other poles and into the ground, or directly into the ground.

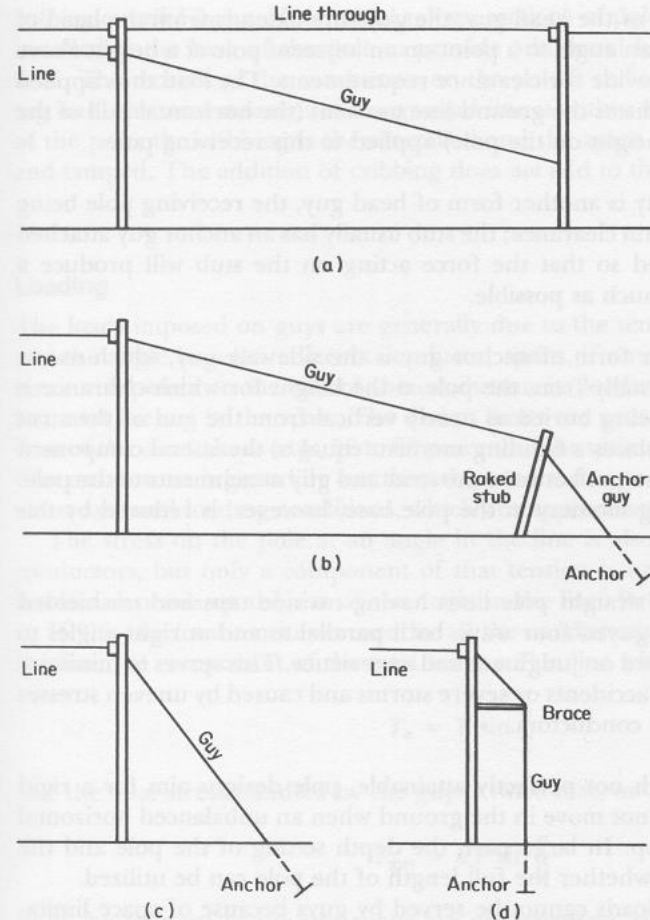


FIG. 5-15 Typical pole guys: (a) pole-to-pole guy; (b) pole-to-stub guy; (c) anchor guy; (d) sidewalk guy.

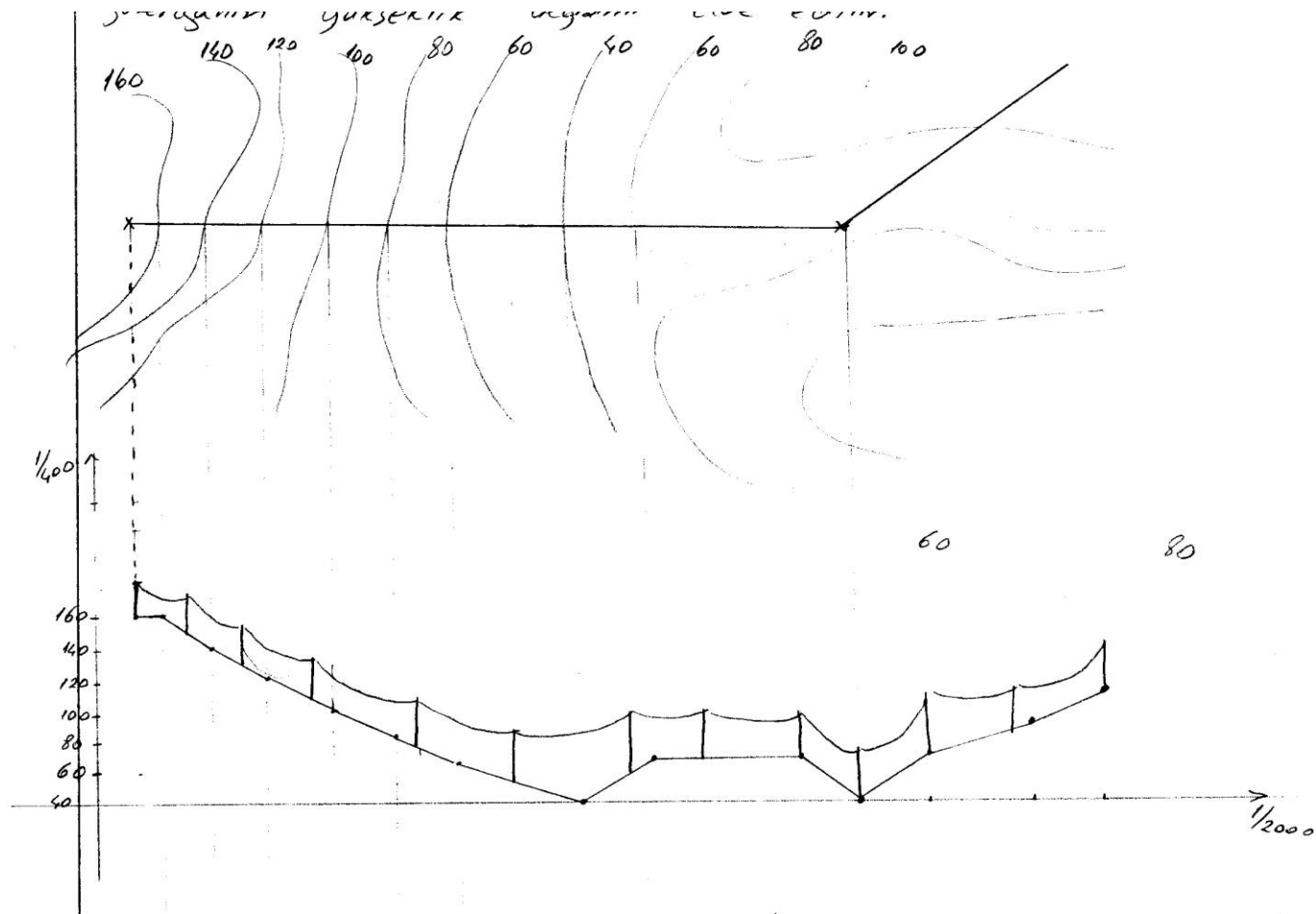
The total loading on a pole is the accumulation of all the individual stress-producing forces that act on the pole. These factors;

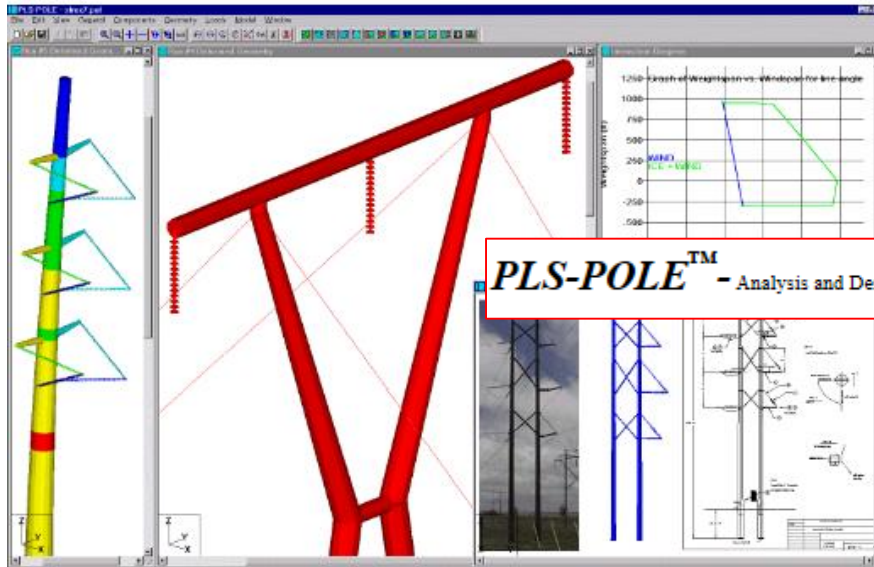
- Tension in wires at dead ends
- Tension in wires due to an angle in the line
- Transverse loading on poles, equipment and wires
- Tension from unbalanced services
- Longitudinal loading

As a practical design method, while a pole line must be designed pole by pole and the necessary strength must be built into each section of the line, much of the repetitious, time-consuming, and tedious calculation can be eliminated through the use of previously prepared tables and charts.

Selecting the pole

- The pole locations and heights are determined by using sag curves (pattern)





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